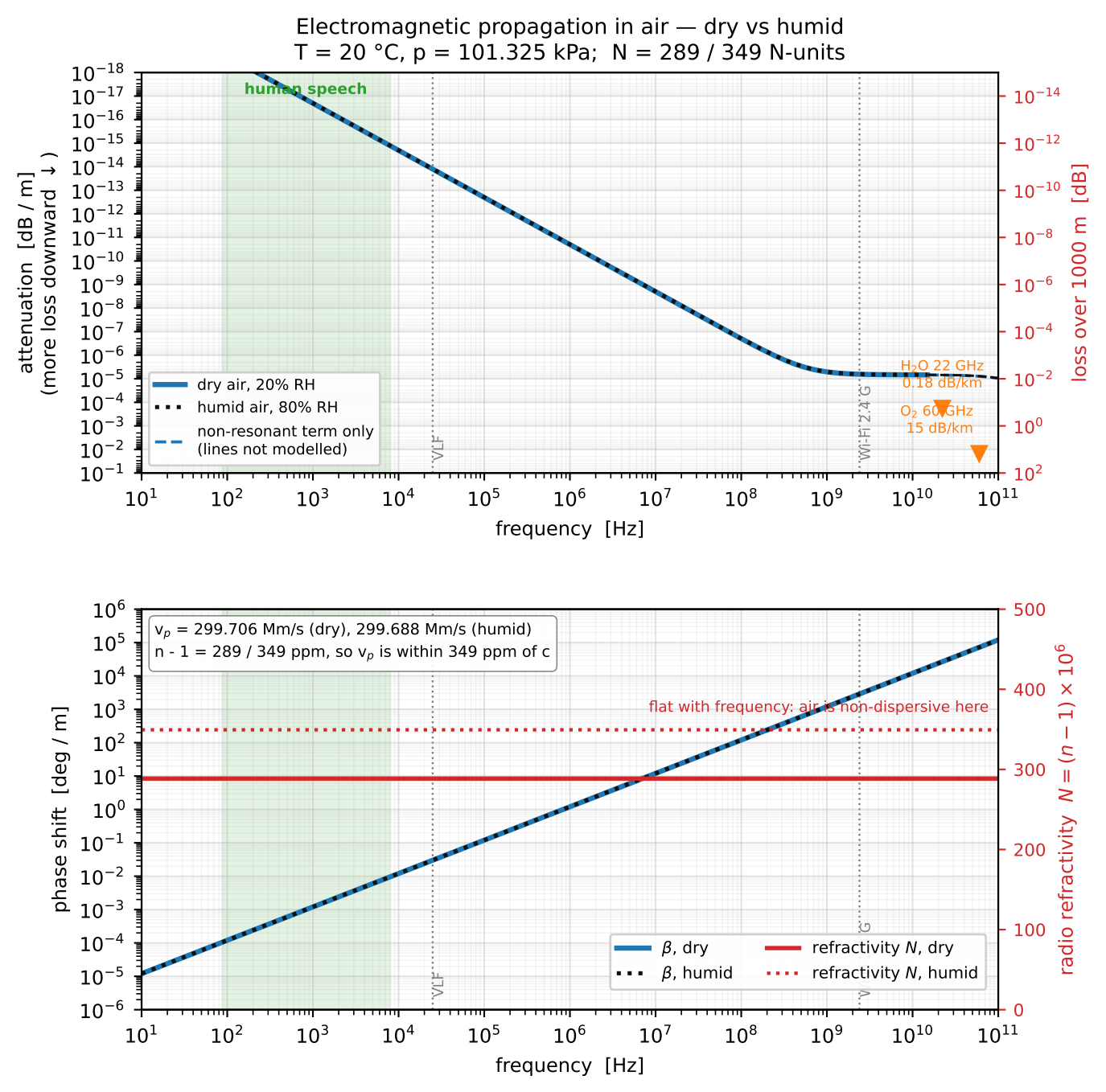
Electromagnetic propagation in air — dry vs humid
T = 20 °C, p = 101.325 kPa; N = 289 / 349 N-units
human speech
VLF
Wi-Fi 2.4 G
H₂O 22 GHz
0.18 dB/km
O₂ 60 GHz
15 dB/km
dry air, 20% RH
humid air, 80% RH
non-resonant term only
(lines not modelled)
attenuation [dB / m]
(more loss downward ↓)
loss over 1000 m [dB]
frequency [Hz]
v_p = 299.706 Mm/s (dry), 299.688 Mm/s (humid)
n - 1 = 289 / 349 ppm, so v_p is within 349 ppm of c
flat with frequency: air is non-dispersive here
β, dry
β, humid
refractivity N, dry
refractivity N, humid
phase shift [deg / m]
radio refractivity N = (n − 1) × 10^6
frequency [Hz]